

SENSOR CONFIGURATION FILE

Autonomous Vehicle System — Team Equinox

aBAJA SAEINDIA 2026

1. Overview

This document provides the complete sensor configuration, embedded hardware integration, vehicle parameter configuration, and autonomous perception architecture used in the Team Equinox autonomous vehicle platform for aBAJA SAEINDIA 2026.

The autonomous stack is designed for real-world deployability using ROS 2 Humble middleware, embedded NVIDIA Jetson compute hardware, and automotive-compatible sensing systems. The architecture integrates:

- Camera-based lane perception (LKA)
- Radar-based obstacle detection (AEB)
- Vehicle state estimation via IMU, encoders, and Hall Effect sensors
- Extended Kalman Filter (EKF) sensor fusion for localisation

All sensor inputs are raw or minimally processed — no ground-truth lane state is used as an algorithm input (compliant with Rule R1).

2. Camera Perception Sensor

The primary perception subsystem for Lane Keeping Assist (LKA) uses a forward-facing RGB camera. Frames are delivered at ~30 FPS and processed through an OpenCV-based pipeline running on ROS 2 Humble nodes, deployed on the NVIDIA Jetson Orin Nano 8GB.

2.1 Camera Specifications

Parameter	Value	Description
Sensor Type	RGB Camera	Primary perception sensor for lane detection
Resolution	640 × 480 px	Optimised for real-time embedded processing
Frame Rate	~30 FPS	Synchronised with the autonomous control loop
Mount Position	Front — Forward Facing	Lane perception and forward navigation
Processing Framework	OpenCV + ROS 2	Computer vision pipeline on Humble middleware
Primary Function	Lane Detection (LKA)	Provides steering correction via PID controller

2.2 Camera Exposure Settings

The default camera parameters produced severely underexposed images (raw pixel values 10–30 / 255, approximately 5 stops below adequate exposure), making it impossible for any classical vision algorithm to distinguish lane markings. The following corrections were applied:

Parameter	Default	Corrected	Rationale
ISO	100	800	+3 stops of sensor sensitivity. Substantially increases brightness without motion artefacts at 30 km/h.
F-Stop	11.0	2.8	+4 stops via wider aperture. Combined with ISO correction, provides sufficient exposure for CV algorithms.

Shutter Speed	250 Hz	250 Hz	Unchanged — motion blur is negligible at 30 km/h.
Tone Mapping	None	Filmic	Compresses HDR → LDR dynamic range. Without this, highlights clip to white even with corrected ISO/aperture.

The apparent night-time rendering was caused entirely by the default exposure/tone-mapping chain, not by scene lighting. Correcting ISO, F-Stop, and tone mapping fully resolved the issue without modifying any environment or sun-angle parameters.

2.3 Perception Pipeline

The camera feed drives a classical 8-stage computer vision pipeline executed at each control tick:

- Image acquisition from Camera RSI stream (TCP, localhost:2210)
- Grayscale conversion
- Binary thresholding
- Bird's-Eye View (BEV) perspective transformation
- Sliding window lane search
- 1st-order polynomial lane fitting
- Lane centre estimation and lateral error computation
- PID steering correction generation

2.4 Camera Mounting & Interface

Parameter	Value
Sensor Interface	Camera RSI (Real Sensor Interface)
TCP Host / Port	localhost : 2210
Output Format	RGB frames over TCP — raw image data, no pre-processing
Field of View	Forward-facing, CarMaker default FoV
Mount Position	Front of vehicle — physically plausible forward-facing position

3. Vehicle State Sensors

The autonomous platform integrates five vehicle state estimation sensors for reliable motion estimation and localisation. All measurements are fused using an Extended Kalman Filter (EKF) to improve accuracy, reject noise, and provide robust dynamic state estimation for the control stack.

Sensor	Type / Technology	Purpose
IMU	Inertial Measurement Unit	Orientation, angular velocity, and linear acceleration estimation
Wheel Speed Sensor	Hall Effect Sensor	Per-wheel rotational speed for slip detection and velocity estimation

Steering Encoder	Absolute Encoder	Real-time steering angle feedback for closed-loop control
Wheel Encoder	Rotary Encoder	Odometry and longitudinal velocity estimation
Motor Controller Sensor	Hall Effect Sensor	Motor rotational feedback integrated into motor controller
Sensor Fusion Layer	Extended Kalman Filter (EKF)	Fuses all above signals for localisation and state estimation

3.1 DVA Quantities Read by Algorithm

The following CarMaker DVA quantities are read at each control tick for vehicle state estimation. These are standard simulator state outputs — they do NOT include any lane-position or road-deviation ground truth (Rule R1 compliant).

DVA Quantity	Unit	Use in Algorithm
Vhcl.v	m/s	Vehicle speed — used to scale Pure Pursuit lookahead distance and PID output gain
Car.YawRate	rad/s	Yaw rate — logged to CSV for post-run analysis and validation
Car.ay	m/s ²	Lateral acceleration — logged to CSV for analysis; not used in control loop

Road Sensor DVA quantities (*Car.Road.Route.DevDist*, *Sensor.Road.Vhcl.VehSensor_1.Route.Deviation.Dist*) are available in the vehicle model but are NOT read by the algorithm — they are reserved for post-run validation benchmarking only and form no part of the control pipeline.

4. Radar Obstacle Detection (AEB)

The Autonomous Emergency Braking (AEB) subsystem integrates an automotive radar sensor for obstacle detection and collision avoidance. The radar provides range, relative velocity, and collision probability data processed through a fuzzy logic controller to determine braking intensity and emergency intervention conditions.

Parameter	Value	Purpose
Sensor Type	Automotive Radar	Primary obstacle detection sensor
Detection Output	Distance + Velocity	Range and relative velocity for collision analysis
Control Algorithm	Fuzzy Logic	Dynamic braking intensity decision under uncertainty
Primary Application	AEB System	Emergency braking and collision avoidance
Operational Benefit	Low-visibility robust	Functions in rain, dust, and low-light conditions

5. Embedded Compute Platform

The full autonomous stack is deployed on an NVIDIA Jetson Orin Nano 8GB — an automotive-grade embedded AI compute module. The software environment integrates ROS 2 Humble for distributed robotics communication, OpenCV for perception pipelines, and simulation tooling for validation.

Parameter	Value	Role / Notes
Compute Platform	NVIDIA Jetson Orin Nano 8GB	Embedded autonomous AI compute module
Operating System	Ubuntu Server (Canonical)	Base OS for ROS 2 deployment
Middleware	ROS 2 Humble	Distributed inter-node robotics communication
Power Supply	19 V Regulated	Vehicle-compatible power input
Simulation Platforms	IPG CarMaker, CARLA, RViz	Validation, testing, and visualisation
Control Frequency	50 Hz	Real-time autonomous control loop rate
Perception Frequency	~30 FPS	Camera processing pipeline rate
EKF Frequency	100 Hz	Sensor fusion update rate

6. Vehicle Model Parameters

All vehicle parameters used in the control algorithm are taken directly from the CarMaker Demo_IPG_CompanyCar physical model specification. No estimated or assumed values are used.

Parameter	Value	Source / Use in Algorithm
Vehicle Model	Demo_IPG_CompanyCar	CarMaker Data/Vehicle/Examples/
Mass	1,391 kg	Vehicle file specification — inertia reference
Wheelbase	2.622 m	Used in Pure Pursuit steering angle computation
Track Width	~1.54 m	Vehicle file specification — lateral reference
Steering Ratio	16 : 1	Maps road-wheel angle to steering wheel — used in Pure Pursuit
Target Speed	30 km/h (8.33 m/s)	Set in config.py; enforced by CarMaker throttle APO channel

7. Compliance with Autonomous Event Rules

The Team Equinox autonomous platform has been designed to comply with all autonomous event requirements specified by aBAJA SAEINDIA 2026. The table below summarises compliance status against each rule.

Rule	Requirement	Status
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R1	No ground-truth object states as algorithm inputs	✓ COMPLIANT — Camera RSI raw frames only; DVA lane quantities not read by algorithm
R2	No simulator-internal APIs exposing privileged world state	✓ COMPLIANT — APO interface used for control outputs only; no scene graph access
R3	Algorithm inputs from configured sensor models only	✓ COMPLIANT — All perception derived from Camera RSI TCP stream
R4	Approved sensor types only (Camera, LiDAR, RADAR, IMU, GNSS)	✓ COMPLIANT — Camera RSI and automotive radar are both approved types
R5	Sensor config validated by Technical Review before presentation	✗ PENDING — Submit this document to Technical Committee
B.1	Real-world reproducibility — no simulation-only shortcuts	✓ COMPLIANT — BEV + Sliding Window pipeline is deployable on embedded hardware
B.5	Min real-time rates: Perception ≥ 10 Hz, Control ≥ 20 Hz	✓ COMPLIANT — Full pipeline runs at 30 Hz in CarMaker